

AC9M6M01 • Year 6 Mathematics

Metric Conversions with Decimal Measurements

Choose units, convert by place value and interpret precision

We are learning to...

Students connect metric prefixes and powers of ten, convert without changing the measured quantity and choose a decimal representation appropriate to the context and instrument precision.

Represent

Reason

Calculate

Interpret

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

convert between common metric units of length, mass and capacity; choose and use decimal representations of metric measurements relevant to the context of a problem

Convert across related metric units

quantity	larger unit	smaller unit
length	2.375 m	2 375 mm
mass	3.48 kg	3 480 g
capacity	1.625 L	1 625 mL
length	0.045 km	45 m

Multiplying or dividing the numeral compensates for changing unit size. A smaller unit needs a larger numeral for the same measurement.

Choose useful decimal precision

road distance

2.35 km more useful than 2 350 m

object thickness

4.6 mm more useful than 0.0046 m

medicine volume

mL and instrument precision matter

shipping mass

kg or tonnes based on scale

Do not report more decimal places than the instrument supports. Preserve exact conversions before rounding for presentation.

Build and annotate the model

Represent metric conversions with decimal measurements and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Numeral changes in the wrong direction:** Smaller units produce larger numerals.
- **Quantity and unit both changed inconsistently:** Use a conversion relationship.
- **Rounding before conversion:** Keep exact value until final presentation.
- **False precision added:** Match instrument capability.

Quick check

- Convert 2.375 m to mm.
- Convert 3.48 kg to g.
- Choose a unit for thickness.
- Explain conversion direction.
- State precision limit.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.